

Pratham Kumar

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PROFESSIONAL SUMMARY

B.Tech Computer Science graduate with a CGPA of 7.63/10 and hands-on experience in full-stack web development using the MERN stack. Skilled in Java, JavaScript, SQL, Data Structures and Algorithms, and REST API development. Solved 120+ DSA problems and built full-stack web applications using MongoDB, Express.js, React.js, and Node.js. IEEE-published researcher with a strong foundation in software development and problem-solving.

TECHNICAL SKILLS

Programming Languages: Java, C, C++, JavaScript, SQL, Python
Web Technologies: HTML, CSS, React.js, Node.js, Express.js, REST APIs, Firebase, MongoDB
Tools & Platforms: Git, GitHub, VS Code, IntelliJ IDEA, Postman, MySQL
Core Concepts: Data Structures & Algorithms, OOP, Database Management Systems (DBMS)

PROJECTS

Image In Kb [GitHub](#)
Full-Stack Developer *React, Node.js, Express, Sharp, MongoDB, Cloudflare Workers, Tailwind CSS*

- Architected a privacy-first, zero-storage image optimization platform delivering sub-100ms transformations using in-memory Node.js buffers, eliminating disk I/O and data retention risk
- Developed an iterative binary-search quality-tuning algorithm to hit exact target file sizes (e.g. 50KB for visa/exam portal uploads), achieving 90%+ size reduction with minimal quality loss
- Deployed globally on Cloudflare Workers with custom SPA routing, automated CI/CD pipeline, and Google Search Console SEO optimization for improved discoverability

BookCart [GitHub](#)
Full-Stack Developer *React, Node.js, Express.js, MongoDB Atlas, Redux Toolkit, Tailwind CSS, Vite, esbuild, Vercel*

- Architected a full-stack MERN e-commerce bookstore platform featuring 350ms debounced search, dynamic price-range filtering (59–199), and a single-page 1-click checkout flow
- Engineered a serverless Express API with global Mongoose connection caching and atomic \$inc stock transactions, eliminating checkout race conditions and reducing cold-start latency by 40%
- Bundled serverless functions into a 65KB standalone distribution using esbuild and deployed on Vercel Edge with SPA rewrites, automated CI/CD, and JWT role-based access control

Real Talks [GitHub](#)
Full-Stack Developer *React, Node.js, Express.js, Socket.IO, MongoDB Atlas, Tailwind CSS, Cloudinary, JWT, Vercel, Render*

- Architected a high-concurrency real-time chat platform delivering sub-10ms bidirectional message delivery via Socket.IO room multiplexing, multi-tab presence tracking, and debounced typing indicators
- Engineered zero-overhead database maintenance using MongoDB native TTL indexes (86,400s) for automated 24-hour message pruning, paired with a dual-layer authentication system (HTTP-Only cookies + Bearer token fallback) to handle cross-origin restrictions
- Developed role-based group channel governance with automated admin succession and a Multer + Cloudinary media pipeline with interactive lightbox zoom, deployed across Vercel Edge CDN and Render

PUBLICATIONS

- "A Systematic Framework for Text-to-Speech System," IEEE, 2025 – Co-authored a research paper presenting a structured framework for TTS system design and implementation

EDUCATION

Sharda University	2022 – 2026
<i>B.Tech in Computer Science</i>	CGPA: 7.63 / 10
Asian School	2021
<i>Senior Secondary Education (Class XII)</i>	Percentage: 84.2%
Asian School	2019
<i>Secondary Education (Class X)</i>	Percentage: 85%

CERTIFICATIONS

- Oracle Academy:** Java Fundamentals, Database Foundations, and AI Foundations Associate – core programming, relational databases, and AI concepts
- Apna College:** Data Structures & Algorithms (DSA) in Java – intensive problem-solving and competitive programming training
- MySirG:** Java Standard Edition (Core Java) – 3-month comprehensive Core Java training program
- MERN Stack Bootcamp** – hands-on full-stack development training using MongoDB, Express.js, React.js, and Node.js

EXTRACURRICULAR ACTIVITIES

- Table Tennis Player:** Represented Sharda University at the All India University Table Tennis Tournament, demonstrating discipline, competitiveness, and time management
- Kartavya NGO:** Active member and facilitator, coordinating community outreach and social impact activities, demonstrating leadership and teamwork